

Global Facilities - Design & Construction Standard - CSA - Concrete Works Specifications

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1.0 Purpose

- 1.1 This document provides a Required Standard and Standard Guideline on Material and Design Specification that shall be followed and complied in the Design, Engineering and Construction for Concrete Works Specifications as part of any green and/or brown field construction projects contract requirement.
- 1.2 The standard is applicable for any green and/or brown field construction projects as per required in the contract.
- 1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.
- 1.4 The established Specifications Standard for each locale shall also comply with Micron's in-house:
 - 1.4.1 RBA Code of Conduct requirements for Semiconductor Manufacturing
 - 1.4.2 Environmental, Health & Safety Standards
 - 1.4.3 Operational Requirements /SOP (Standard Operation Procedures)
 - 1.4.4 Security Protocol
- 1.5 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws and reflects the current semi-conductor industry's best practices.

- 1.6 Where there is the option to adopt either the MasterSpec or in-region specifications standard, the AE (Architectural & Engineering Consultants) shall propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.

2.0 Scope

- 2.1 This document for concrete works specification covers all necessary related design, engineering, procurement, construction, installation and completion of piling, substructure, superstructure, and roof structure to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.
- 2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

- 3.1 Definitions of acronyms used in this document are listed as follows:

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
AE	External Architectural & Engineering Consultants
DPT	Micron's Designated Project Team
GFCT	Micron's Global Facilities Construction Team
EHS	Micron's Environmental Health & Safety Team
QP/PE	External Qualified Person (professional architect/engineer)
SME	Micron's Subject Matter Expert
TM	Micron's Team Members

4.0 General

Materials used in the concrete works shall be of the qualities and kinds specified herein and equal to approved samples. Delivery shall be made sufficiently in advance to enable samples to be taken and tested if required. No materials shall be used until and unless approved and materials not approved shall be immediately removed from the works at the Contractor's expense.

Materials shall be transported, handled and stored on the site or elsewhere in such a manner as to prevent damage, deterioration or contamination all to the satisfaction of the Engineer.

Where the requirements of the relevant Local Code Standards are in conflict with this Specification then this Specification shall take precedence.

Contractor shall submit to the Engineer the appropriate certificate of compliance with the Standards. Notwithstanding the above, tests shall be carried out as and when directed by the Engineer in accordance with the relevant Standards.

5.0 Cement

Unless otherwise specified or instructed by the Engineer, cement shall be of an approved brand and manufacture and comply with the following standards or equivalent local code requirements:

<u>Type</u>	<u>Relevant Standard</u>
Ordinary Portland Cement	SS EN SS 197-1
Rapid Hardening Portland Cement	SS EN SS 197-1

High alumina cement shall not be used for any concrete work.

Before starting any work the Contractor shall submit for approval the type and brand of cement he proposes to use. The Contractor shall not change the cement brand unless he has specific reasons to do so and permission in writing shall be obtained from the Engineer

Contractor shall produce a certificate showing the date and place of manufacture of the cement. As and when directed by the Engineer, samples of cement shall be provided for testing. Any cement failing to comply with the requirements of the relevant standard shall be rejected.

The manufacturer's test certificate will normally be accepted as proof of compliance with the Specification but the Engineer reserves the right to carry out further tests in an approved Accredited Laboratory.

The Engineer may without tests being made order any bags of cement a portion of the contents of which has hardened or which appear to be defective in any way to be removed from the site at the Contractor's expense.

The compression test and not the tensile test shall be used to determine the strength of cement.

The cement shall be fresh and delivered in unbroken bags of the manufacturer or in properly designed bulk containers. All cement delivered in bags shall be stored in a

weatherproof dry shed with raised boarded floor of at least 300mm above natural ground level.

Cement delivered in bulk tankers shall be stored in properly designed bulk storage tanks.

Cement of different manufacture and types shall be kept separate at all times and shall not be used in the same mix.

Concrete of different brands of cement shall not be mixed during casting. The Engineer shall be informed in writing of any change in the source of supply.

Consignments of cement shall be used in the order of delivery and any caked or lumpy cement shall not be used. All cement unfit for use shall be removed from site immediately.

Each consignment shall be identifiable and the Contractor shall supply without cost to the Engineer a copy of Manufacturer's Test Certificate for each consignment.

The age of the cement at the time of delivery to the site shall not be more than 2 months and the cement shall be used within 3 months of delivery.

6.0 Aggregates

Materials used as aggregate shall be obtained from sources known to produce aggregate satisfactory for concrete work.

Materials used as aggregate shall be chemically inert strong, hard, durable, of limited porosity and free from adhering coatings, clay lumps, coal or coal residue and organic or other impurities that may cause corrosion of the reinforcement or impair the strength or durability of the concrete.

Fine aggregates shall be natural sand or crushed gravel sand or crushed stone sand complying with SS EN 12620 or local code requirements.

Coarse aggregates shall be crushed or uncrushed gravel, crushed or uncrushed stone complying with SS EN 12620 or local code requirements.

Coarse aggregates shall not exceed 40mm in size for lean and mass concrete and 20mm for reinforced concrete, unless otherwise specified.

All-in aggregates as specified in SS EN 12620 or local code requirements shall not be used.

Aggregate used shall not be alkali reactive.

Aggregates delivered to site shall be stored on clean surfaces in separate stockpiles or bunkers to prevent intermingling of different materials and contamination.

The Engineer reserves the right to draw samples of aggregates from any stockpile at any time and have them tested in accordance with and BS 812 for compliance with SS EN 12620 or local code requirements. Aggregates failing to comply with SS EN 12620 or local code requirements shall be rejected.

Aggregate whose grading is found to fall outside the approved zone shall be rejected.

The Engineer shall be informed in writing of any change in the source of supply of these

aggregates.

7.0 Water

Potable water supplied by the Public Utilities Board shall be used for mixing concrete, mortar and grout. For cleaning of formwork, reinforcement and curing of concrete, water other than potable water may be used if approved by the Engineer. The use of seawater shall be prohibited.

Contractor shall arrange for tests of the water to be carried out in accordance with EN 1008 or local code requirements.

8.0 Admixtures

When admixture is proposed to be used in concrete mix, furnish information concerning the admixture, admixture acceptance tests and additional information as stipulated in SS EN 934 and SS EN 480 or local code requirements for approval by the Engineer prior to the use of admixture.

Admixture used shall comply with SS EN 934 and SS EN 480 or relevant or local code requirements. Requests for permission to use other types of admixtures not covered by these standards may only be considered when accompanied by full supporting information. Admixture to be used shall be approved by the Engineers.

The Contractor shall provide the Engineer every facility for inspection and sampling of the admixture(s) at least once every three-month of active concreting period. The sample(s) shall be kept for reference and shall be sent, if requested by the Engineer, for admixture uniformity tests in accordance with SS EN 934 and SS EN 480 or the relevant or local/international code requirements

9.0 Concrete Mixes

9.1 The requirements for materials and concrete mixes shall be in accordance with SS EN 206 and SS EN 544 or local code requirements.

9.2 Prescribed Mix

Unless otherwise stated, concrete mix shall be Designed Mix. However, prescribed mix may be used provided :

- (a) the work is of minor nature or involving a small quantity of concrete;
- (b) approval is given by the S.O.;
- (c) the strength of the concrete is still the responsibility of the Contractor.

With prescribed mix the concrete mix shall be as tabulated in **Table 1**. The mix design in the following table does not require the use of admixtures.

Table 1: prescribed mix the concrete mix

Concrete Grade:	Workability	Medium	High
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C25/30	Limits to slump that may be expected (mm)	25 - 75	75 - 125
	Cement (kg)	380	410
	Total Aggregated (kg)	1735	1670
	Fine Aggregate * (% by weight of total aggregate)	47	47
	Water (litre)	210	225

9.3 Design Mix

For designed mixes, the Contractor shall submit a mix design report covering all concrete mixes to the S.O. for approval. The designed mix shall comply with the following requirements in **Table 2**:

Table 2: Designed Mix requirements

Concrete Grade	C15/ 20	C20/ 25	C25/ 30	C28/ 35	C32/ 40	C35/ 45	C40/ 50
Min. Cement Content (kg/m ³)	205	260	275	300	325	350	400
Max. Cement Content (kg/m ³)			550	550	550	550	550
Max. Fine Aggregate (%)			50	50	50	50	50
Total Aggregate							
Max. Water/Cement Ratio	0.7	0.6	0.55	0.5	0.45	0.40	0.40

The minimum cement content shall be increased by 20 and 40 kg/m³ respectively for concrete with maximum aggregate size of 14 and 10mm.

Mix design shall be based on a target mean strength not less than the specified characteristic strength plus 8 N/mm².

In all cases, it is the responsibility of the Contractor to ensure that the designed mix shall be durable, workable and comply with the strength grade requirement as specified in the Drawings.

Design of mix shall ensure no segregation, no excessive bleeding and ability to achieve specified surface finish. Segregation and bleeding shall be determined by visual inspection.

9.4 Trial Mixes

The designed mix with a designed maximum consistency as measured by slump or flow shall be verified by preparation of trial mixes witnessed by the S.O. representative. The mix shall be judged for suitable workability as well as strength. A minimum of 12 cubes from three batches shall be taken, six of which shall be tested at 7 days and the other six at 28 days.

The mix may be considered suitable if :

- (a) the designed maximum consistency is achieved in accordance with EN 12350, EN 12390, EN 12504 and BS 1881 or local code requirements if a flow test is specified ;
- (b) the average strength of the six cubes tested at 28 days exceeds the target mean strength.

No concrete shall be placed in the work until the results of the trial mix are approved by the S.O.

All the trial mixes and cube tests shall be carried out at the expense of the Contractor.

Trial mixes may not be required if concrete is supplied by ready-mix concrete suppliers approved by the S.O.

9.5 Mix Adjustment

Once the designed mix is approved by the S.O., the contractor shall not be allowed to change the mix without prior approval of the S.O.

An increase or reduction of cement content by 20 kg/m³ and above, or % fine aggregate by 5% and above, or any changes in the type or dosage of admixture or consistency is considered a change in mix design. This does not constitute a tolerance allowance on the mix design used in relation to the specified requirements.

The S.O. reserves the right to instruct the Contractor to modify and improve the mix design from time to time during the progress of work as and when he considers necessary to meet the requirements. This shall not entitle the Contractor to any extra payment.

10.0 Batching

Cement shall be measured by weight or batched in whole bags if so packed and delivered. Fine and coarse aggregates shall be measured by weight separately. Water shall be measured by weight and due allowance shall be made for the amount of water present in the mass of fine aggregate by deducting this amount from the weight of water to be added during the mixing of concrete. Solid admixtures shall be measured by weight, while liquid or paste admixtures may be measured by weight or volume.

The accuracy of the measuring equipment shall be within $\pm 3\%$ of the quantity of cement, water or total aggregate being measured, and with $\pm 5\%$ of the quantity of admixture added before commencement of work and subsequently whenever so directed by the Engineer, the measuring equipment shall be calibrated with known weights and volume measures.

11.0 Mixing Concrete

Concrete shall be mixed in a batch type mechanical mixer. The quantity of constituent material in each batch shall not exceed the rated capacity of the mixer. The constituent material shall be thoroughly and continually mixed at the mixing speed specified by the mixer manufacturer until the resulting concrete is uniform in colour and

consistency.

The mixing time shall be as follows :

- (a) Pan mixer : minimum 30 seconds.
- (b) Truck mixer or drum mixer : minimum 1.5 minutes for 1m³ of concrete plus ¼ minute for each additional 1m³.

The mixing time shall be reckoned from the time when all the solid materials are in the mixer. No portion of the time required for discharging shall be considered as part of the mixing time.

The first batch of concrete fed into the mixer shall have an increase of an appropriate amount of dry mortar similar in proportion to that of the concrete to be mixed and of mixing water for coating the mixer.

The mixer shall be emptied completely before being refilled. Mixers which have been out of use for more than 30 minutes should be thoroughly cleaned before any fresh concrete is mixed. The mixer shall be free from hardened concrete. Worn-off blades or paddles shall be replaced.

No concrete shall be mixed by hand except with the specific approval of the Engineer for a small amount. Where such approval is granted, the Contractor shall at his own expense increase the proportion of cement by 10%. Mixing shall be continued until the concrete is of uniform colour and consistency.

12.0 Ready-Mix Concrete

Where ready-mix concrete is used, it shall comply with SS EN 206 and SS EN 544 or local code requirements and the mix design and admixture used shall be approved by the Engineer. The cement content shall comply with Clause 6 'Concrete Mixes'.

No water in excess of the quantity required in the approved mix shall be allowed to be added to the concrete to increase its workability affected by elapsed time and/or temperature. While it is not being discharged, the concrete shall be kept continuously agitated. The concrete shall be placed in its final position and left undisturbed within two hours from the time when the cement is added to the mix.

The Contractor shall furnish the name of his ready- mixed concrete supplier to the Engineer for approval. He shall make such arrangements as may be required for the Engineer to inspect his supplier's work and provide all facilities for samples of cement, fine and coarse aggregates and admixture to be taken for testing.

Notwithstanding any such inspection made and/or approval given by the Engineer, the Contractor shall take full responsibility for ensuring that all ready- mixed concrete supplied shall conform to SS EN 206 and SS EN 544 or local code requirements and this Specification. Concrete that does not comply with this Specification shall be rejected and removed from site.

The Contractor shall furnish to and obtain the approval of the Engineer at least 1 day in advance of the date set for each casting of concrete, the number of transit trucks/truck mixers/truck agitators proposed to be used to supply the concrete and the frequency at which the loads of concrete are to be delivered to the site.

For each truck load of concrete delivered the following information shall appear on the delivery docket :

- name of ready-mixed concrete batch plant
- serial number of docket
- date and the licence number of the truck
- name of the purchaser
- name and location of job
- specified characteristic concrete strength and the amount of cement used (kg) or the mix proportions
- quantity of concrete
- time of loading or of first mixing of cement and aggregate, whichever is earlier
- agreed slump
- maximum size of coarse aggregates specified
- trade name of admixture, if any
- the signature of the Engineer, who shall also enter the time of arrival of the truck and the time of completion of discharge
- position where concrete is placed
- number of cubes taken, if any, and cube references
- any other requirements as may be instructed

One copy of the docket shall be given to the Engineer on delivery.

The Engineer reserves the right to instruct the Contractor to change the supplier due to unsatisfactory performance or to rescind his approval for further use of ready-mixed concrete during the progress of the works, if any of the requirements has, in his opinion, not been satisfactorily complied with.

13.0 Workability and Consistency

The workability shall be such that the concrete can be well compacted, worked into the corners of the shuttering and around the reinforcements, gives the specified surface finish and attain the specified strength. The mix design shall ensure that this workability is obtainable.

On each concreting day, workability shall be assessed on site at the discharge point of the mixer by means of the slump test in accordance with SS 12350, EN 12390 and EN 12504 or local code requirements. If the measured slump deviates from the agreed value by more than $\pm 25\text{mm}$ or one third of the required value whichever is greater, the concrete shall be rejected.

As an alternative, flow test to BS 1881 : Part 105 or other means of assessing workability may be considered subject to approval by the Engineer.

No water shall be added to the concrete subsequent to the completion of the mixing.

The required workability for any part of the work shall be determined by the Contractor, subject to the agreement of the Engineer. It shall be reviewed during the progress of the work and adjustments shall be made as necessary subject to the agreement of the Engineer.

14.0 Distribution of Concrete

14.1 General

The concrete shall be distributed from the mixer to the position of placing in the works by approved means which do not cause segregation or otherwise impair the quality of the concrete.

Details of any proposed pumping conveyor belt or chuting plant must be approved by the Engineer before such plant is delivered to the site.

All distributing equipment shall be cleaned before commencing mixing and distribution of the concrete and such equipment shall be kept free from set concrete.

14.2 Concrete Pumps and Pipelines

The rate of supply shall match with the rate of pumping.

Prior to commencement of pumping, check must be carried out to ensure that the feeder hopper and the pipelines are free from obstruction, the coupling pipe joints are properly tightened and the pipeline is firmly secured and anchored. Pipeline and its support must not be rigidly connected to formwork.

Pumping and the agitator inside the feeder hopper shall normally be operated in a continuous manner. When a stoppage is unavoidable, the concrete inside the hopper must be maintained at a level high enough to prevent any entry of air into the pipeline to form an air pocket. For temporary stoppage, the concrete in the pipeline shall be moved occasionally by a few strokes of the piston of the pump, thus preventing it from sticking.

Pumping shall be stopped in the following occasions:

- presence of foreign matters which may block the valves or pipelines
- level of concrete in the hopper is too low
- an unsuitable batch of concrete is placed in the hopper
- extra lengths of pipe are to be added

Foreign matter and unsuitable batch of concrete must be removed immediately from the hopper before pumping is resumed.

15.0 Placing Concrete

No concrete shall be placed prior to the Engineer's approval.

All concrete shall be transported from the mixer to the place of final deposit as speedily as possible before it has commenced to set, and in no case shall this exceed 20 minutes after mixing. Where retarders are used such interval may be extended subject to the approval of the Engineer. The method of transit shall be such that it will prevent the segregation, loss or contamination of the ingredients.

In the use of concrete pumps, the Contractor shall ensure that the following is complied with:

- (a) Pipes shall not be secured to formwork or reinforcement which may be displaced by the pulsating movement of the pipe.
- (b) Pumping shall commence first with blowing of water through the pipeline, followed by cement mortar, both of sufficient quantities and all these shall be discharged away from the area to be concreted.
- (c) When extra sections of pipe are to be added, each length of pipe shall be thoroughly wetted inside but no free water shall remain in the pipe.
- (d) Where the pipeline is exposed to the sun, the line shall be protected against absorption of heat by covering with damp sacking or other approved material during pumping.

Before proceeding to place the concrete, the formwork shall be re-aligned if necessary and water and rubbish therein shall be removed. Immediately prior to placing the concrete, the formwork shall be well wetted and inspection openings shall be closed.

Raised gangways for workmen placing and vibrating concrete shall be provided wherever possible and shall be so constructed as to avoid damaging and displacing the reinforcement.

Concrete shall be placed in the formwork by shovels or other approved equipment and shall not be dropped from a height nor handled in a manner which will cause segregation. Accumulations of set concrete on the reinforcement shall be avoided. Concrete shall be placed directly in the permanent position and shall not be worked along the formwork to that position.

For beams and similar members, concrete shall be placed in a single operation to the full depth. In walls, columns and similar members, the concrete shall be placed in horizontal layers each not exceeding 1m deep. Placing of concrete shall be continuous and temporary interruption shall not normally exceed 30 minutes. At the completion of a specified part, a construction joint shall be made subject to the approval of the Engineer.

Any water accumulating on the surface of newly placed concrete shall be removed by approved means and no further concrete shall be placed thereon until such water has been removed.

The output of concrete shall be sufficient to construct each section between predetermined construction joints in one operation. Any construction joint not shown on the Drawings is to be approved by the Engineer before casting.

In the event of unavoidable stoppage in positions not predetermined the concreting shall be terminated on horizontal planes and against vertical surfaces by the use of stopping off boards.

No concrete shall be placed during wet weather, unless the entire area of pour is covered, or other measures are taken to protect the wet concrete, to the satisfaction and subject to the approval of the Engineer.

All concrete shall be placed in daylight, unless an adequate system of lighting is provided and approval for night work is given by the Engineer and all relevant authorities prior to commencement of the concreting.

A record shall be kept at the Site Office of the time and date of placing the concrete in each portion of the structure and the reference numbers of the test cubes pertaining to batches of concrete for each particular part of the structure.

16.0 Compaction of Concrete

All concrete shall be compacted to produce a dense homogeneous mass. It shall be compacted using approved immersion type mechanical vibrators. Sufficient numbers of vibrators in serviceable condition shall be used on site to compact the concrete at the rate at which it is placed, and a minimum of two standby vibrators shall be readily available for emergency use in case of breakdowns.

The concrete shall be thoroughly worked into all parts of the formwork and between and around the steel reinforcement. It shall be compacted to give a dense and compact concrete, free from voids of any kind. Steel reinforcement shall be prevented from being displaced or deformed during concreting.

Internal immersion type vibrators shall be used in all sections which are sufficiently large to permit their use.

For slabs, screen-vibrators shall be used in conjunction with internal vibrators if the depth is such that the latter may be used.

Form vibrator shall be used when sections are too small for the internal type.

Vibration, with any type of vibrator, shall not be continued in any one spot to the extent that pools of grout are formed. Care shall be taken to avoid any disturbance of concrete which has become too stiff to regain plasticity when vibrated.

Vibration shall not be used as a means of distributing heaped concrete into position.

After being compacted and finished, the concrete must be left absolutely undisturbed while setting.

17.0 Curing and Protection of Concrete

After being placed in position, all fresh concrete shall be protected from rain, sun and drying wind, and shall not be subjected to any vibrational disturbance other than that arising from compacting. Exposed surfaces of concrete shall be kept moist and covered for a minimum of 3 days after placing, with a material such as polythene sheet or a curing compound or an absorbent material kept constantly damp by frequent watering. The method of curing and the covering material used shall be approved by the Engineer.

Provision shall be made for adequate water distribution to all parts of the works so that if required curing can be maintained efficiently throughout the whole period of construction.

The use of proprietary curing compounds shall be subjected to the Engineer's approval. Where such approval has been granted the curing compounds shall be applied in strict accordance with manufacturer's instructions.

Concrete placed below ground level shall be protected from falling earth and vegetation, and the excavated ground shall be kept free of any water by pumping or other means of drainage. Approved means shall also be taken to protect immature concrete from damage by debris, excessive loading, moving vehicles, mixing with earth or other materials, flotation and other influences that would impair the strength and durability of the concrete. Concrete placed in ground containing deleterious substances

shall be free from contact with such ground and with water draining therefrom, during placing and for a period of 3 days or as otherwise instructed thereafter.

Curing of freshly placed concrete shall commence immediately after trowelling and shall continue without interruption for a period of not less than 3 days.

For floors that are not to be trowelled, curing shall commence as soon as the surface will not be damaged by the curing process.

All materials and equipment necessary for the curing of concrete must be available and ready for use before concrete arrives.

For 'concrete finished' surfaces, the type of curing method used shall be such that wrinkles or marks will not appear on the finished surfaces.

Curing freshly placed concrete by ponding with water shall be carried out such that adjacent areas are kept dry.

If concrete is cured by covering the surface with wet sacks, these must be kept continuously wet over the curing period. Sacks shall overlap at the edges and weighted down to prevent lifting by strong wind.

Polythene or plastic sheets used for curing shall be at least 0.20mm thick and shall be clear or white in colour.

Curing compounds shall not be used on concrete surfaces which are to be covered with concrete or other finishes unless it can be shown that the compound used will not weaken the bond between the surface and the covering.

Curing compound shall be sufficiently thick to form an effective film soon after application.

No traffic shall be applied to the surface during the curing period.

Concrete damaged through improper or insufficient curing and protection shall be made good at the Contractor's own costs and expenses.

18.0 Hot Weather Concreting

18.1 When placing concrete in hot weather, concreting shall be rapidly placed and effectively compacted and continuity of construction maintained.

Once the concrete has been compacted into place, appropriate curing measures shall take place to confine the water within the concrete and avoid evaporation of moisture in the concrete and prevent thermal and plastic cracking. Such measures may take the form of either or a combination of temporary impervious sheeting with sprayed curing compounds, water-ponding and covering with damp materials when the concrete surface is sufficiently hard.

The subgrade and formwork shall be sprinkled with water before concreting. All pools of water formed through excessive sprinkling shall be removed.

In important structural members and in exceptionally hot weather and drying winds, the Contractor may be required to confine concreting to the cooler parts of the day.

- 18.2 The following measures to the ingredients of the concrete shall be taken to reduce the initial temperature of concrete once it reaches 40°C and above :

Aggregate stockpiles shall be sprinkled with water regularly (especially in layers as placed) and shaded. Aggregate may be sprayed also with cold water immediately before use.

Water pipelines may be treated to ensure that no rise in the temperature of water occurs prior to its being used. Pipelines may be buried, insulated or shaded. If necessary, the mixing water may be cooled with ice.

Cement in bags shall be shaded. Cement stored in silos shall be protected from heat from sunlight. Silos shall be shaded against radiation or painted white or other reflective paint externally.

Mixers, chutes, forms etc. shall be shaded where practicable.

- 18.3 The temperature of concrete at the point of delivery shall not exceed the prevailing shade temperature by more than 6°C.

19.0 Concreting of Thick Sections

- 19.1 In massive pours in raft foundations and concrete sections thicker than 600mm, the Contractor shall take all necessary precautions and measures according to proven practices in concrete technology to prevent any form of cracking due to temperature and shrinkage effects in the concreting.

Cement with low heat of hydration such as Low Heat Blastfurnace Slag Cement or Low Heat Portland Cement shall be used in conjunction with other methods of temperature control in concrete to prevent development of temperature gradients sufficient to cause thermal cracking.

The temperature gradient between any two points within any concrete section shall not exceed 20°C at anytime. The Contractor shall propose his method of thermal curing or any other measures to comply with this requirements.

The Contractor shall provide and install thermocouples and all necessary equipment and instruments at locations to be decided by the Engineer to measure and monitor the concrete temperatures. The Contractor shall propose effective remedial actions to the Engineer for approval should the temperature gradient exceeds the specified limit at any time.

Notwithstanding the above measures and precautions, the Contractor shall seal any cracks found on any concrete elements with approved epoxy resin at his own expenses.

- 19.2 Trial mix of concrete using the low-heat cement proposed shall be carried out together with thermal instrumentation on a 1.5 metre cube concrete mass to measure the temperature gradients and differential developed.

20.0 Quality of Concrete Works

20.1 General

Concrete shall be tested for compliance in strength and durability requirements.

The compressive strength of concrete shall be determined by crushing tests of 150mm cubes. The quality and consistency of the concrete shall be tested as directed by the Engineer in accordance with SS 78. The Contractor shall provide for slump test or flow test, compressive strength test, core test and any other concrete test as and when required by the Engineer.

20.2 Sampling & Testing

All sampling, curing and testing of cubes shall be carried out in accordance with EN 12350, EN 12390 and EN 12504 or local code requirements. A sample shall be taken from a randomly selected batch of concrete chosen by the Engineer.

On each concreting day, for both precast and cast-in- situ concrete, for every 40 cubic metres or part thereof, of each grade of concrete per concrete mixer, at least one sample shall be taken from a batch chosen by the Engineer. A sample shall be a quantity of concrete, consisting of a number of standard scoopfuls, taken from a batch of concrete whose properties shall be determined. Two cubes shall be cast per sample per age of testing unless specified otherwise. The sample shall be taken at the point of discharge from the mixer or agitating truck of a wet batching plant, or, in the case of a dry batching plant, at the point of discharge from the mixer truck.

At least four cubes shall be cast from each sample. Two cubes shall be tested at an early age of 3 days for precast concrete and 7 days for cast-in-situ concrete. The remaining two cubes shall be tested at 28 days for compliance purpose.

On completion of the curing, the specimens shall be tested and the mean of the two results shall be taken as the test result. When the difference between the two results divided by their mean exceeds 15%, the test result shall be deemed invalid.

For precast concrete, additional cubes shall be taken from the same sample to determine whether adequate strength is achieved in order to meet design strength for demoulding, handling and delivery. These additional cubes shall be cured by the same methods as those used for the precast units. Testing facilities approved by the Engineer shall be provided at precast plant for testing of these additional cubes.

In addition to the minimum number of samples stated above, the Engineer may, at his absolute discretion, instruct the Contractor to take extra samples from whichever batch as he may choose.

The Contractor shall provide suitable equipment for the measurement of concrete cover to reinforcement and in- situ strength of the concrete. The Contractor shall carry out measurements for all major structural elements as determined by the Engineer. The Contractor shall analyse statistically all measurements for each of the major type of structural element. The concrete cover shall comply with the requirements of SS EN 1992 or local code

requirements. The in-situ strength of the concrete shall be assessed in accordance with EN 13791 or local code requirements where applicable. The strength shall comply with the specified grade, taking into account the variability with in-situ strength.

The cost for sampling, making, curing, forwarding the cubes for testing and measuring of concrete cover and in-situ strength shall be borne by the Contractor. The cubes for testing shall be sent to an Accredited testing laboratory approved by the Engineer. Testing fees will be borne by the Contractor. Should the Engineer so require, the Contractor shall provide facilities for making specimens for Flexural Strength Tests in accordance with SS 78 or local code requirements.

20.3 Compliance of Strength Requirement

For cast-in-situ concrete, the 7 days strength shall be required for quality control purpose as well as for compliance with the strength specified below. This is to ensure adequate strength to withstand the construction load.

For precast concrete, the 3 days strength shall be obtained solely for quality control purpose. Early strength requirements to meet design strength for demoulding, handling and delivery shall be specified by the approved manufacturer of the precast components and approved by the Engineer. Such strength shall be determined from the additional cubes. The Engineer shall have the absolute right to reject any precast components should the early strength fail to meet the design strength requirements.

Compliance with the specified characteristic strength shall be judged by tests made on cubes at the age of 7 and 28 days. Compliance may be assumed if both the following conditions are satisfied:

(i) The average strength of the sample and the preceeding three samples or the average of the first four samples of the production is greater than or equal to the specified characteristic strength plus half a fixed current margin of 6 N/mm². Thus, the minimum average strength shall be:

Grade		C25 /30	C28 /35	C32 /40	C35 /45	C40 /50
Minimum Average	7 days	24	28	32	36	40
Cube Strength (N/mm ²)	28 days	33	38	43	48	53

All test cubes taken shall be properly stored and the Contractor shall ensure that

(ii) The sample cube strength is at least the level specified below :

Grade		C25 /30	C28 /35	C32 /40	C35 /45	C40 /50
Minimum Individual	7 days	18	22	27	30	34
Sample Strength (N/mm ²)	28 days	27	32	37	42	47

all test cubes are tested at their respective age. Test cube that are lost shall be deemed to have failed.

20.4 Action to be taken in the Event of Non-compliance

If the sample strength results do not comply with either or both of the above conditions the Contractor shall redesign his mix based on an increased target mean strength. The section of work represented by the samples which fail the test shall be removed and replaced by concrete meeting this Specification. The Engineer may direct that other measures be taken to make the works secure. The Engineer may instruct that additional tests be carried out on the hardened concrete.

The affected concrete work may be stopped and not continued until so directed by the Engineer. No extension to the contract period shall be allowed for such stoppage of the work and remedial works.

All works instructed under this sub-clause shall be at the Contractor's expense.

20.5 Cubes Tested at 7 Days

When the 7-days strength is less than that stipulated in Clause 17.3, no more concrete shall be placed on the suspect concrete until the 28-days strength is found to comply.

Any additional expense and delay shall be borne solely by the Contractor.

21.0 Testing of Hardened Concrete

In the event of non-compliance, the quality of concrete in question shall be interpreted in accordance with SS EN 206.

For 7 days non-compliance, prolonged propping shall be provided until the 28 days strength becomes available.

For 28 days non-compliance, the quality of the sub- standard concrete shall be estimated by carrying out tests on hardened concrete in the structure. These may include non-destructive methods, the taking of core samples, load tests or combination of several methods as follows :

Grade	Average Compressive Cube Strength (N/mm ²)	Test on Structure After 28 days
C25/30	27.0 to below 33.0	
C28/35	32.0 to below 38.0	non-destructive
C32/40	37.0 to below 43.0	<u>method</u> and/or
C35/45	42.0 to below 48.0	<u>core</u> test
C40/50	47.0 to below 53.0	
C25/30	Below 27.0	
C28/35	Below 32.0	non-destructive method
C32/40	Below 37.0	<u>and</u> core test and/or
C35/45	Below 42.0	<u>load</u> test
C40/50	Below 47.0	

The test shall be carried out by an accredited laboratory in accordance with SS EN 1992 and assessed in accordance with EN 13791. The results of any such tests shall not nullify the establishment of non-compliance provided that the latter are based on valid cube tests results. Notwithstanding the outcome of such tests, the Contractor shall bear all the costs and expenses incurred for such tests.

21.1 Core Test

The Engineer reserves the right to direct that samples be cored from any portion of works for the purpose of examination and testing.

The cutting equipment and method of coring to obtain specimens shall be approved by the Engineer.

The specimens shall be dealt in accordance with BS 1881. Prior to preparation for testing, the specimens shall be made available for examination by the Engineer.

The cored portion of the works shall be made good to the satisfaction of the Engineer.

21.2 Check Test

Consequent upon work cube test failure, the Engineer may instruct that other appropriate tests be carried out in accordance with SS EN 1992 or local code requirements on any portion of the works for the purpose of assessing the strength. The Contractor shall bear all the cost for such test.

22.0 Structural Tests

22.1 General

The Engineer may instruct the Contractor to make a load test on the works or any part thereof if in the Engineer's opinion such a test is necessary.

The Contractor shall bear all the costs and expenses incurred for such test.

22.2 Load Test

i) Building Works

For building works the load test shall be done generally in accordance with SS EN 1992.

ii) Precast Units

Testing of the precast units shall be carried out as and when directed by the Engineer in accordance with SS EN 1992 or local code requirements. Samples of the precast unit for the tests shall be provided and delivered by the Contractor at his cost to an approved laboratory.

Where more than one sample from any sample batch of two or more similar units fail to satisfy this Specification, then the entire batch of units represented by such samples shall be liable for rejection by the Engineer.

For precast concrete members which shall subsequently form parts of composite

members, any of the following methods may be used at the discretion of the Engineer.

- (a) The precast member may be tested along with a load system equivalent to that producing the most critical conditions in the composite members.
- (b) The precast member may be made equivalent to the final composite member by casting the relevant amount of in-situ concrete, and then tested as a composite member.

22.3 Test Results

If the results of any of the test is not satisfactory, at the Engineer's instruction, that part of the works shall be rejected and reconstructed to comply with this Specification, all at the Contractor's expense. No extension of time shall be entertained in this connection.

23.0 Rectification of Defective Works

The action to be taken in respect of the concrete which fails to meet either of the requirements in Clause 17 'Quality of Concrete Works' and the assessment specified in Clause 18 'Testing of Hardened Concrete' shall be determined by the Engineer. They may range from:

- (a) Qualified acceptance in less severe cases. The Contractor shall provide protective coating against carbonation and other aggressive environment if the affected concrete can be certified to be structurally sound by a Professional Engineer engaged by the Contractor at its own costs and expenses; or

If the average of three core tests falls below the strength level specified in Clause 17.3(ii), or in the event of failure of a load test.

- (b) The Contractor shall carry out at his own costs and expenses any necessary approved remedial measures to the satisfaction of the Engineer. Remedial measures shall be designed, supervised and certified by a Professional Engineer engaged by the Contractor at its own costs and expenses; or
- (c) The Contractor shall remove the affected concrete, and the costs and expenses in removing all poor quality concrete shall be the responsibility of the Contractor.

All defective concrete work such as badly constructed concrete members with poor alignment and plumb, honeycombed and badly formed surfaces shall be removed or otherwise rectified according to the instructions of the Engineer at the Contractor's own costs and expenses.

24.0 Steel Reinforcement

24.1 Description

Unless otherwise shown on the Drawings, reinforcement for concrete shall consist of plain mild steel and high yield reinforcement bars complying with SS 560 and welded steel fabric complying with SS 561 or local code requirements. Their respective specified characteristic strength shall not be less than 250 N/mm², 460 N/mm² and 485 N/mm².

Reinforcement shall be free from pitting, loose rust, mill scale, paint, oil, grease, adhering earth or any other material that may impair the bond between the concrete and the reinforcement or that may cause corrosion of the reinforcement or disintegration of the concrete.

Where reinforcement is to be welded, it shall comply with SS EN 1992 or local code requirements. The Contractor shall obtain the approval of the Engineer in writing prior to any welding of reinforcement.

24.2 Testing

The Contractor shall provide the Engineer every facility for inspection and sampling before any cutting, bending or other fabrication is undertaken. If deemed necessary, the Engineer may cause samples to be taken from reinforcement already fixed in position.

Steel reinforcement shall be tested by an Accredited Laboratory approved by the Engineer. Reinforcement not complying with the appropriate specifications shall be removed and replaced at the Contractor's own costs and expenses.

The reinforcement shall be tested in accordance with the relevant Singapore Standard. Certificates of origin and manufacturer's test certificates shall be produced. Selection of test pieces shall be made by the Engineer. The Contractor shall bear all costs involved in making these tests. The Contractor shall have no claim for bars mutilated in obtaining the test pieces.

The Contractor shall furnish, for reinforcement supplied by him, manufacturers' certificates and these shall be submitted for approval by the Engineer before any materials is brought on to the site.

The Contractor shall also submit test record issued by approved laboratory giving the test results of the mechanical properties including tensile strength, yield point and elongation.

24.3 Galvanized Steel

Galvanized mild steel and galvanized high yield steel reinforcement bars as indicated in the Drawings shall be hot-dip galvanized to EN ISO 1460 and EN ISO 1461 or local code requirements. The Contractor shall obtain galvanizing service or galvanised steel bars only from approved galvanisers or suppliers.

The coating weight of zinc shall conform with EN ISO 1460 and EN ISO 1461 or local code requirements. For steel over 5mm in diameter, the minimum average coating mass shall be 500g per m² and the minimum coating mass on any individual sample shall be 450g per m². The coating shall be continuous, smooth and free from flux stains. Should the coating appearance of any galvanised reinforcement bar be found otherwise, that bar shall not be used as a galvanised reinforcement bar. Galvanising shall be followed by chromating as a post

treatment. Store all galvanised reinforcement bars in a dry and well ventilated place to prevent the formation of white rust on these bars. Small areas of galvanised coating damaged by welding, cutting or by excessively rough treatment shall be made good by the application of at least two coats of good quality zinc-rich paint with minimum 85% of zinc dust, expressed as a percentage by weight of the solid content of the paint.

Upon the instruction of the Engineer, the Contractor shall send at least two samples of the galvanised reinforcement bars to be tested by a competent authority to determine the coating weight of these bars. The coating weight shall be determined by stripping tests in accordance with EN ISO 1460 and EN ISO 1461 or local code requirements. Certificates of such tests shall be submitted to the Engineer for verification.

25.0 Fabrication Reinforcement

All reinforcement shall be accurately cut and bent to the dimensions and shapes shown in the Drawings, and in accordance with EN ISO 3766 or local code requirements.

Bends shall be cold-formed by machine or by other approved means which produce a gradual and even bending action. Reinforcement incorrectly bent shall not be used unless the method for straightening and rebending will not cause any cracking, and approval for rebending is given by the Engineer.

Reinforcement shall be securely tied with steel wires. The projecting ends of the steel wires shall be directed away from the formwork so as not to encroach into the concrete cover.

Reinforcement shall be adequately supported by approved spacers and chairs to maintain the specified concrete cover. Plastic spacers shall be used for all bottom slab reinforcement. Steel bar chairs bent according to the Drawings or proprietary-type chairs shall be used for top slab reinforcement. Parts of the chairs that need to rest on formwork shall be coated with plastic for isolation and protection. All spacers and chairs for slab reinforcement shall be placed not more than 600mm apart in a regular grid.

Other than for slabs, both plastic and concrete or mortar spacer blocks shall be provided with tying wire, and shall not be lower in compressive strength and durability characteristics than the surrounding concrete. All spacers shall be obtained from sources approved by the Engineer.

Sleeve jointing for reinforcement may be used provided that the strength and deformation characteristics have been determined by tests carried out by a competent testing authority and approved by the Engineer.

Fabrication and placing of reinforcement in each part of the work shall be approved by the Engineer before any concrete can be placed thereon.

Reinforcement shall not be displaced during concreting. Reinforcement projecting from work being concreted or already concreted shall not be bent without approval by the Engineer and shall be protected from accidental deformation and damage.

Butt jointing of reinforcement bars by welding may be permitted under suitable conditions and with suitable safeguard in accordance with EN 1011.

Welding shall not be done unless approved by the Engineer. All welding shall be done by appropriately skilled welder.

The reinforcement shall be thoroughly wire-brushed and cleaned to remove loose rust mill scale paint, oil grease adhering earth or any other material that may impair the bond between the concrete and the reinforcement or that may cause corrosion of the reinforcement or disintegration of the concrete before placing the bars into position.

Unless otherwise specified in the drawings, minimum cover to reinforcement shall be as shown in Table below. Nominal cover is measured from the concrete surface to the outermost reinforcement including links. In any case, the cover shall not be less than the diameter of the bar and for bundle bars, the cover shall not be less than the size of a single bar of equivalent area.

Concrete Grade Nominal Cover	C25/30 mm	C28/35 mm	C32/40 mm	C35/45 mm	C40/50 and over mm
Concrete above ground in buildings	25	20	20	20	20
Buried concrete and concrete continuously underwater	40	35	30	25	20

26.0 Formworks

26.1 Material

All formwork shall comply with SS CP 580 or local code requirements and shall be, in every respect, rigidly constructed of approved material and shall be true to the shape and dimensions as shown in the Drawings. Timber used for formwork shall be medium hardwood, machined dressed to a uniform thickness. The timber shall be well seasoned, free from loose knots and, unless otherwise approved, wrought on all faces.

Phenolic film faced plywood, complying with SS 1 or local code requirements and cut to size from sheets at least 13mm thick, shall be used to produce a good off-form finish. The form face in contact with concrete shall be free of dirt, adhering grout, projecting nails, splits and other defects.

Formwork shall be coated with an approved form oil which is compatible with the required finish. Such oil shall be insoluble in water, non-staining, not injurious to concrete and shall not become flaky or be removed by rain or wash water. Diesel oil and liquids which retard the setting of concrete shall not be used. The same approved type of form oil shall be used for all work. The application of the form oil shall be such that no reinforcement is coated by the oil and there shall be no accumulation of the oil on the formwork.

26.2 Design

Formwork shall be designed to SS CP 580 or local code requirements and shall cater for the appropriate method of placing and compacting the concrete. Adequate number of braces, ties and struts shall be provided so that the formwork shall be able to withstand the pressures resulting from whatever

method of placing and compacting concrete adopted, and maintain the required position, slope and profile.

Metal props shall be used to support formwork, except at irregular places where timber props may be more suitable or practical. Use of timber props shall be subject to approval by the Engineer.

Where the floor to ceiling height exceeds 9.0 metres, or where the formwork deck is supported by shores constructed in two or more tiers, or where the dead, live and impact loads on the formwork exceed 730.0 kg/per square metre, the Contractor shall engage a Professional Engineer at its own costs and expenses to design the formwork structure. The Contractor shall submit the details of the design of the formwork and the supports for approval by the Engineer. Notwithstanding the approval by the Engineer the strength and adequacy of the formwork and supports shall remain the responsibility of the Contractor and his Professional Engineer.

26.3 Construction

Formwork shall be constructed sufficiently rigid and tight to prevent loss of grout from the concrete at all stages of placing and compacting.

The arrangement of formwork shall be such that it can be dismantled and removed readily from the cast concrete without causing shock, disturbance or damage. For beams and slabs the arrangement shall be such that the formwork and supports for the slab soffits and beam sides can be removed without disturbing the formwork and supports of the beam soffits.

Supports of an upper floor shall rest on the lower floor only if the lower floor has developed adequate strength to bear the load. If in the opinion of the Engineer, the lower floor does not have the required strength to carry the support of the upper floors, then the method adopted to support the formwork of the upper floor shall have to be approved by the Engineer.

Connections in formwork shall be either nailed, screwed, bolted, clamped or otherwise secured. Ties that are anchored against the timber shall pass through a plate washer at least 50mm x 50mm in size and of sufficient thickness so that the load is transferred to the timber without any visible deflection or penetration into the timber. All holes left in the concrete shall be filled with mortar and finished to the approval of the Engineer.

The Contractor shall be required to make allowances for any settlement or deflection of formwork which is likely to arise during construction. The soffits of beams and slabs shall be given a camber to allow for deflection after striking of formwork. The magnitude of the camber shall be approved by the Engineer.

Unless otherwise shown on the drawings, the soffits of beams and slabs shall be given an upward camber towards the centre of the span.

Element	Camber	
	Span > 5m and up to 10m	Span > 10m
Beams	0.2% span	0.3% span
Slabs	0.2% span	0.3% span
	Cantilever span > 1m	
Cantilevers (beams or slabs)	0.4% span	

For flat slab construction, use the 'diagonal' span between supports. The cambers to which flat slabs are laid shall be agreed with the Engineer prior to work commencing.

Formwork across structure joints shall be erected such that it overlaps adjoining finished concrete by at least 200mm and holds tightly against it. Formwork shall be provided and run continuously at construction joints.

All formwork joints shall be sufficiently tight to prevent leakage of cement grout and to avoid the formation of fine or other blemishes and all faulty joints shall be caulked. Where the appearance of the concrete face is important the position and direction of the joints shall be as directed.

Bolts to be permanently installed in concrete, boxouts, cores and other devices used for forming openings, holes, pockets, recesses and other cavities shall be as rigid as the formwork itself to prevent the leakage of any grout. No holes shall be cut in cast concrete except with the approval of the Engineer.

If the Contractor fails to provide for any boxout, hole, opening or insert which he is expected to ascertain from services drawings and to co-ordinate with all sub-contractors, and if the Engineer considers that cutting the concrete to provide such opening may impair the strength of the member, then, the contractor shall at his expense adopt an approved method which the Engineer may deem fit to provide for such opening.

Formwork for Slopping Surfaces :

Top shuttering shall be provided to concrete faces where the slope exceeds 1 to 2.5. Such shuttering shall be counter-weighted or anchored against flotation.

All rubbish, chippings, shavings etc. shall be removed from the interior of formwork immediately before concrete is placed, first by compressed air and then by a jet of water. Temporary openings shall be formed for the inspection of the inside of the formwork and for the escape of water used for washing.

All formwork shall be inspected by the Engineer after preparation and immediately prior to depositing concrete and no concrete shall be deposited until approval of the formwork and reinforcement has been obtained from the Engineer.

26.4 Sample Panels

Before the commencement of work on concrete to be exposed permanently, the Contract shall erect a sample of the formwork and cast the sample finish. The finish as approved by the Engineer must be maintained throughout the Contract. Where the resulting finish of a completed work is not considered by the Engineer to meet the standard of the approved sample panel the Contractor may be required to remove the work completely and reconstruct or carry out such surface treatment as may be required by the Engineer. This work shall be done entirely at the Contractor's expense. For concrete to receive an applied finish, the Contractor shall prepare sample panels to assess the self finishing properties of the chosen mix and its compatibility with the applied finish.

27.0 Striking of Formworks

Formwork supporting cast-in-situ concrete shall not be removed unless :-

- (i) such removal will not result in unacceptable deflections due to shrinkage, creep, etc.
- (ii) such removal has been accounted for in the sequence of construction which has been previously agreed upon between the Contractor and the Engineer.

The minimum periods before striking of the formwork for ordinary portland cement concrete shall be according to the following table :

Type of Formwork	Min. Period Before Striking	Remark
Vertical formwork to columns, walls and side of beams	12 hours	The period shall commence from the time of last pour of concrete and shall run the full days until it lapses, before striking of formwork can begin. Time shall be counted on hourly basis taking a full day as 24 hours.
Soffit formwork to slabs and beams	3 days	
Props to slabs and beams	14 days	After striking of formwork, props, shall be <u>immediately</u> put up to support the structure.
Props to cantilever slabs and cantilever beams	21 days	

Shorter striking periods for beams and slabs may be permitted provided evidence is provided to the Engineer that the strength development in the in-situ concrete is such that unacceptable deflections and cracking will not occur as a result of earlier striking. No formwork or support shall be struck or removed without approval by the Engineer. Formwork shall be removed gradually without shock or vibration and in the manner and order approved by the Engineer.

The Contractor shall record on the drawings or elsewhere the date upon which the concrete is placed in each part of the work and the date upon which the formwork is removed.

After striking of formwork, any honeycombed surface and defective or damaged concrete work arising from improper work procedures shall be immediately made good as directed by the Engineer and at the Contractor's own costs and expenses. Any remedial measures carried out without the knowledge and approval of the Engineer shall be liable for rejection and re-execution.

All holes formed by tie bolts shall be filled with cement mortar as specified in Clause 33 'Cement Mortar and Grout'.

All formwork shall be removed without damage to the concrete. The Contractor shall be responsible for any injury to the work and any consequential damage caused by or arising from the method adopted for the striking or moving of the formwork or supports.

28.0 Opening and Recesses

Provide openings and recesses in the reinforced concrete structure, including PVC sleeves where required, for the installation of sanitary pipes and fittings, service ducts or pipes, CATV, TAS, and electrical cables and trunkings, as shown in the Drawings. The Contractor shall leave openings in the lift motor room floor to the requirement of the lift contractor or as instructed by the Engineer. No openings shall be formed without the prior approval of the Engineer.

No coring of holes through existing structural elements shall be permitted. However, holes not exceeding 20mm in diameter may be drilled through these elements. If any steel bar obstructs the drilling through of a hole, a new position shall be selected slightly offset to avoid the obstructing bar. Under no circumstances shall any steel bar be allowed to be cut or removed without the written consent of the Engineer. For holes with diameter greater than 20mm, submit drilling proposals to the Engineer for approval.

All openings in slabs shall be formed with sides tapering slightly downwards and trimmed on all sides with steel bars as shown in the Drawings. Check and ensure accuracy in the position, alignment size and shape of all openings and recesses. Otherwise, the Contractor shall make the necessary rectification at his own costs and expenses.

Remove all formwork other than PVC sleeves required from openings and recesses after the concrete has set, and ensure that the openings are not blocked up. Recesses for vertical electrical trunkings shall be finished smooth.

Unless otherwise indicated in the Drawings, the Contractor shall seal openings in floor slabs with similar grade of concrete or with cement mortar as specified in Clause 33 'Cement Mortar and Grout' both with approved water proofing admixture added, after installation of services by the Contractor or other Contractors. Other openings without PVC sleeves shall be made good with cement mortar as specified in Clause 33 'Cement Mortar and Grout'.

29.0 Concrete Surfaces

As soon as formwork has been removed and after inspection by the Engineer, honey-combed surfaces shall be epoxy pressure grouted. Small holes in the surfaces shall be filled with cement mortar as specified in Clause 33, 'Cement Mortar and Grout'. Concrete surfaces shall be treated as follows except stated otherwise elsewhere in this specification.

- (i) all projecting imperfections shall be grinded with carborundum stone or other means approved by the Engineer and the grit or dirt thoroughly washed off with clean water.
The contractor shall grind the surface when "Green" immediately after the formwork have been removed, all honey-combs and holes repaired, leaving a smooth and even surface.
- (ii) as a separate operation after completion of (i) above surfaces shall be brushed over with a coating of neat Portland Cement wash which shall be rubbed into the pores and smoothed off with carborundum blocks.

30.0 Building Accuracy

After removal of formwork, the Contractor shall take measurements as directed by the Engineer to check the deviation of the reinforced concrete work from specified dimensions shown in the Drawings and the current NPQS (National Productivity & Quality Specifications).

All measurements shall be recorded and submitted to the Engineer. Finished concrete within the tolerances specified in the Drawings and the NPQS shall be deemed acceptable.

Unless specifically stated, otherwise in the Drawings or NPQS, permitted tolerances shall be in accordance with the current issue of BS 5606 – Guide to Accuracy in Building or local code requirements.

Where there are different criteria or values for tolerances stated in the Standard (BS 5606) and Codes of Practice (SS CP 580 – Formwork and SS EN 1992) or local code requirements, the most onerous shall apply.

31.0 Construction Joints

The position and details of construction joints proposed by Contractor shall be subject to the approval of the Engineer and shall be arranged so as to minimise the possibility of occurrence of shrinkage cracks. The Contractor shall submit for approval drawings showing the intended sequence of pouring, location, and details of such construction joints. No casting shall commence until approval has been received from the Engineer.

Construction joints shall be at right angles to the general direction of the member and shall take due account of shear and other stresses. Joints shall be rebated and of an approved shape and shall be vertical or horizontal as required. There shall be minimum construction joints to all wet-areas (toilets, wash- areas/yards, balconies, kitchen, roof and basement). Where joints occur, these shall be made waterproofed to the Engineer's approval.

Concrete in a beam shall be placed throughout without a joint, but if the provision is unavoidable, the joint shall be vertical and at $\frac{1}{4}$ to $\frac{1}{3}$ of a span. Joint in slab shall be

vertical and parallel to the principal reinforcement and where it is unavoidable, at right angles to the principal reinforcement, the joint shall be vertical and at $\frac{1}{4}$ to $\frac{1}{3}$ of the span. Concrete in a cantilever-member shall be placed throughout without a joint.

If a kicker is used, it shall be at least 70mm high and shall preferably be incorporated in the previous concrete.

Waterstops shall be provided for all horizontal and all vertical construction joints of liquid and/or earth retaining structures, (including basement wall and slab, lift-pits, swimming pools and balancing tanks). Waterstops shall be installed and butt jointed according to BS 8007 or local code requirements and the Manufacturer's recommendations. Rearguard PVC waterstops 250mm wide x 3mm thick shall be provided for floor joints and vertical wall joints and kicker joint.

The cost of providing the waterstops is deemed to be included in the Contractor's price, excluding those indicated on drawings.

Whenever the placing of concrete is interrupted due to inclement weather, the concrete must on no account be allowed to tail off at random but is to be stopped and thoroughly compacted against properly constructed stopping-off-boards to form an efficient construction joint.

Before concreting resumes, the joints shall be properly chipped, cleaned and scum and loose aggregate removed therefrom. Immediately before placing new concrete, the joint shall be thoroughly wetted and coated with rich cement grout.

The Contractor shall submit progressively for the Engineer's record, plans showing as-constructed locations of construction joints and the dates on which adjacent concrete panels were cast.

32.0 Permanent Structure Joints

32.1 General

Expansion joints, contraction joints or other permanent structure joints shall be provided in the positions and constructed and sealed with waterproofing materials as detailed in the Drawings.

All exposed vertical sides of columns and walls and undersides of beams at expansion joints shall be covered with 0.7mm thick aluminum cover strips fixed with masonry nails at 300mm c/c or otherwise as shown in the Drawings.

The installation method and the selection, mixing, application and curing of all joint waterproofing materials shall comply with the Drawings and the manufacturers' recommendation. Alternative joint waterproofing materials may be used subject to the approval of the Engineer. Submit supporting technical information, test reports and samples of the proposed waterproofing materials.

The joint and its supporting area shall be protected from loading and disturbance until they have reached adequate strength. This shall be done by providing and maintaining suitable ramps which shall not be removed until permitted by the Engineer.

All waterproofing materials used at public access areas shall be protected with

non-shrink grout.

32.2 Waterstops

Waterstops shall be installed and butt jointed according to SS EN 1992 - 3 or local code requirements and the Manufacturers' recommendations. Waterstops shall be securely positioned in the formwork to prevent displacement during concreting.

32.3 Sealing Compound

Sealing compound for external use shall be one of the three generic types namely polyurethane, polysulphide and silicone. One-part polysulphide and one-part polyurethane shall comply with EN ISO 11600 or local code requirements. Two-part polysulphide and two-part polyurethane shall comply with EN ISO 11600. Silicone sealant shall conform to EN ISO 11600 and also satisfy the 'Adhesion after heat again test'.

32.4 Preformed Flexible Strip Sealant

Preformed flexible strip sealant shall comply with the following requirements :-

- (a) It shall only be used in horizontal joint and be subject to compression throughout its length;
- (b) Good adhesion;
- (c) Water resistant;
- (d) Non-staining

32.4 Bitumen/Rubber Cold Applied Membrane

Bitumen/rubber cold applied membrane shall comply with the following requirements:

- (a) Min. joint movement capacity of $\pm 10\%$ of joint width at 25°C;
- (b) Resistant to dilute acids and alkali.

32.5 Hot-Poured Rubber/Bitumen Sealing Compound

Hot-poured rubber/bitumen sealing compound shall comply with BS 2449.

32.6 Bituminous Sheet

Bituminous sheet with fibre shall comply with the following requirements :-

- (a) Resistant to lime water (no visual effect after two weeks immersion);
- (b) Max. water absorption of 10% of dry weight;
- (c) Min. tensile strength of 50 kg/cm²;
- (d) Ozone and ultra-violet resistance.

The strip shall be applied to the joints by burning and affixing on each side of joint to a width not less than 70mm leaving the middle part free to take any movement at the joint without damage.

32.7 Neoprene Bearing Pads

Neoprene bearing pads shall comply with the following requirements :-

- (a) Shore 'A' hardness of 60 ± 5 at 25°C ;
- (b) Min. rupture strength of 105 kg/cm^2 ;
- (c) Min. rupture elongation of 300%.

32.8 Polyethylene Foam Backing Rods

Polyethylene foam backing rods used as mortar stops in panel joints shall have the following properties :-

- (a) Min. compressibility of 75% of original volume at 25°C ;
- (b) Excellent resilient properties;
- (c) Density between 35 kg/m^3 and 45 kg/m^3 .
- (d) Totally resistant to common acids, lubricants and detergents;
- (e) Totally resistant to water infiltration by capillary action;
- (f) Suitable for up to 70°C ;

33.0 Apparatus

The Contractor shall provide the following apparatus for use for all Works on site :-

- (a) Concrete slump test apparatus complying with EN 12350, EN 12390 and EN 12504 or flow test apparatus complying with BS 1881: Part 105 or local code requirements. One set of the apparatus shall be provided for each concreting location.
- (b) Steel or cast iron moulds complete with rod in accordance with EN 12350, EN 12390 and EN 12504 or local code requirements for casting 150mm concrete test cubes and 100mm mortar or grout test cubes. A minimum number shall be provided such that no stripping of cube is required prior to 24 hours setting and hardening period.
- (c) Three measuring cylinders of 250ml capacity, graduated to measure to the nearest 2.0ml, for determination of silt content (field settling method).
- (d) An approved apparatus for measuring moisture content in fine aggregate.
- (e) One concrete rebound hammer.
- (f) One surveying level or the odolite complete with the necessary accessories.
- (g) One electronic calculator with statistical functions.
- (h) One 5.5m measuring tape;
- (i) One spirit level; and
- (j) A thermometer complying with the requirements of BS 1704 or local code requirements.

The above apparatus shall be provided at the commencement of the Contract. The Contractor shall be responsible for the safe keeping and cleaning of the apparatus after each use. No concreting shall be allowed unless the above apparatus have been provided for immediate use.

34.0 Curing Tanks for Concrete Test Cubes

For the storage of normal test cubes, the Contractor shall provide lockable water tight curing tank(s) of adequate size to store all test cubes. The tank(s) shall be placed under a covered area or at a shaded spot. The keys to such curing facility shall be exclusively held in custody by the Engineer.

35.0 Lean Concrete

Lean concrete shall be laid on well prepared firm ground before any reinforcement is placed in position.

Immediately following the excavation to formation level for floors, beams, or foundations a blinding coat shall be placed on the surface of the formation and carefully spade finished to an even surface. Utmost care shall be taken to preserve the natural water content of the ground below formation level. Floors or foundations shall be laid on the blinding coat as soon as may be practicable, this being especially important during hot weather. If required by the Engineer, the blinding coat shall be kept thoroughly damp by watering.

Unless otherwise stated, a 50mm thick lean concrete layer shall be provided on all soil surfaces forming the underside of any reinforced concrete beam and slab, footing and pile caps, composition of the lean concrete shall be Grade C10/15 concrete.

36.0 Cement Mortar and Grout

36.1 General

For structural purposes, cement mortar shall compose of one part cement to one part sand, mixed with water so that water-cement ratio does not exceed 0.4 by weight and cement grout shall compose of cement and water- cement ratio between 0.4 and 0.6, or such other proportions as shall be directed by the Engineer.

36.2 Precast Concrete Joints

Non-load bearing joints between precast concrete components and adjoining structures shall be filled with appropriate grout and/or mortar protected by proprietary sealants and backing rod. They shall be waterproof.

Load bearing joints and connection shall be grouted, mortar packed or concreted. The respective mix design shall be free of lime and chloride. They shall be durable, waterproof, non-shrink and possess strength equal or higher than that of precast concrete. Curing for at least 3 days shall be provided. Designed mixes shall be submitted to the Engineer for approval.

36.3 Testing Method & Compliance for Load Bearing Grout & Mortar

The method of sampling and testing prescribed in BS 4551 shall be adopted. The compressive strength shall be determined by crushing test on 100mm cubes. For each casting day and for each grade of grout or mortar, three samples shall be taken from three separate batches. One cube shall be cast from each sample for testing at 28 days.

The compliance of the 28 day strength shall be judged, based on the average compressive strength of 3 cubes from 3 samples taken on the same day, in accordance with the strength requirement specified in the Drawings.

37.0 Bonding Tie-Bars

6mm diameter mild steel bonding bars shall be fully welded onto 25mm wide x 2.3mm thick mild steel plate ramset to reinforced concrete columns, walls and stiffeners where hollow block or solid block or brick partitions abut for bonding. The 6mm diameter mild steel bars shall be welded onto the mild steel plate at every fourth course for bonding with projection not less than 300mm into the partitions. Welding of bars shall be 50mm full welding to both sides of the bar onto the plate. The mild steel plate shall be of length as shown and shall be set with 3.8mm diameter, 28mm long masonry drive pin to the reinforced concrete columns, walls and stiffeners at spacing of 400mm. The depth of penetrations of masonry drive pins shall be minimum 25mm

38.0 Foundations

All reinforced concrete footings and pile caps shall be constructed in accordance with the Drawings and to the exact depths required. Provide, maintain and remove any necessary planking and strutting, sheet-piling or cofferdams, and by pumping or other approved means keep the excavations free from water.

In the case of loose or disturbed ground, the bottom of excavation for pile caps shall be well rammed and those for footings shall be excavated further to a sound layer. The whole work shall be approved by the Engineer before it is covered with a blinding layer not less than 50mm thick. The binding layer to be used for pile caps and footings is lean concrete of Grade 15 concrete. The required cover of concrete under the reinforcement shall be entirely above the blinding layer.

The Contractor shall allow in his contract for time necessary for extension and/or cutting-off/stripping of piles by the piling contractor.

He shall provide all labour/machineries for excavation to enable the piling contractor to carry out such extension and/or cutting-off works.

39.0 Pile Caps

Before commencing to construct pile caps, the Contractor shall check and verify the eccentricities and the cut-off levels of all piling work in the ground against pile eccentricity drawings provided and pile cap details in the Drawings, and shall notify the Engineer in the event of any discrepancy.

Straighten the steel reinforcement projecting above the piles for anchoring pile caps, carry out excavations, erect formwork and temporary timbering for the construction of pile caps.

40.0 Lift Pits

For lift pits below ground level, the concreting to the bottom and walls shall be carried out in one operation without construction joints and bolt holes up to the underside of ground beams and slabs.

All external wall surfaces of lift pits shall be waterproofed with three coats of bitumen solution complying with BS 3416 or other cold applied bituminous coating as approved by the Engineer. The concrete surface shall be clean and dry before application of the first coat, and ample time for each coat to dry shall be allowed. The dried total thickness of the coating shall be 0.6mm minimum.

41.0 Refuse Chutes

All cast-in-situ reinforced concrete refuse chutes shall be constructed without bolt holes made through the chute walls. If such holes are unavoidable then they shall be completely grouted with cement mortar as specified in Clause 32. 'Cement Mortar and Grout' or sealed and waterproofed by other approved means to prevent leakage, to the satisfaction of the Engineer.

The Contractor shall allow for casting refuse chute frames into refuse chute walls for fixing chute hoppers. Such frames shall be obtained from the same supplier of aluminum hoppers.

42.0 Columns / Piers and Walls

Before the column form is placed (and lined up and plumbed) the concrete of the base is to be bush hammered until the aggregate over the whole contact surface is exposed.

The column form is to have across the bottom of one long side a 75mm high opening and suitable closing board. One side of the form is to be opened from 1.25m upwards by removing suitable closing pieces first fitted and then removed.

Before concreting, the form and especially the bottom is to be well watered outside and inside and cleaned by flushing. Concreting is to start by pushing in a 50mm layer of grout through the bottom hole and this hole is thereafter to be closed. A bucket of grout is now to be tipped into the column which is then to be concreted.

43.0 Floors

43.1 Concrete to receive Applied Finish

When the concrete is to receive an applied finish, the formwork shall be made of closely jointed wrought boards. Alternatively, steel or other materials may also be used, subject to the Engineer's agreement. On removal of the formwork and upon Engineer's inspection, defects shall be treated. The concrete face shall then be roughened by approved means to form a key before the finish is applied.

For areas which are to receive rendering or other finishes, the fresh concrete shall be struck off to a level or fall as required using a long timber trowel. The surface shall be densified by a jitterbug tamper or trowelling after the bleeding

has stopped in order to eliminate settlement and plastic shrinkage cracks. Before it hardens, it shall be brushed with a stiff broom in one direction to give a rough and tidy surface.

43.2 Slab Finish Without Topping

After initial set, the upper surface of cast-in-situ reinforce concrete floors shall be trowelled smooth with a steel float to true level and even surface.

The floating shall be done so as not to bring an excess of mortar to the surface of the concrete. Indentation in the surface of the concrete shall be formed by approved implements to the depths and patterns described.

Care shall be taken to ensure that the steel reinforcement is not displaced or lowered during trowelling.

43.3 Slabs to receive Topping

Just prior to placing the topping, the base slab shall be thoroughly cleaned, prepared for bond and wetted. No pools of water shall be left standing on the wetted surface. A thin coat of neat cement grout shall be applied onto the surface of the slab for a short distance ahead of the topping. The wearing surface shall be applied before the grout has hardened.

43.4 Permanently Exposed Concrete Finishes Derived From Formwork

The formwork for permanently exposed concrete shall be specially designed to provide the required finish. The faces of the formwork against which the concrete is to be placed shall be hard and with a wrought finish or lined with an approved non-staining form lining if required. An approved non-staining form oil shall be used to coat the face to facilitate easy removal of the formwork. The entire formwork shall be sufficiently rigid to give clean true finish.

Before commencement of work, sample panels shall be erected as provided in Clause 23.4. The number of reuses permitted for timber formwork shall be subject to approval of the Engineer.

The texture and quality of the surface finish shall be consistently maintained throughout the project. The type of cement, formwork materials, form oil and the source of the aggregate shall remain unchanged at all stages of the project. The Contractor shall, at the commencement of the work, ensure that adequate supplies of these materials are available for the duration of the construction.

During the placing of the concrete, the Contractor shall take special care to thoroughly compact the concrete especially in areas nearer the formwork in order that the finished surface is free of water and air holes.

After the formwork is carefully removed, the Contractor shall notify the Engineer who shall inspect the surface. If instructed by the Engineer, the Contractor shall then proceed to remove any fins and wash down the surface.

Thereafter, only the most noticeable surface blemishes shall be filled with a cement and fine aggregate paste, the colour of which shall match the colour of the concrete.

All finished work shall be very well protected from damage, staining, discolouration and any other defect at all stages of construction.

43.5 Watertightness and Water Stagnation Test

After the completion of all structural work on the floors, all toilets, bathrooms, wash-areas, balconies and terraces' slabs shall be tested for watertightness and ensure no water stagnation.

The watertightness test shall be carried out by ponding with water to a minimum depth of 25mm for 24 hours. At the end of the watertightness test, the soffit of the slab especially the area around the pipes shall be examined for any dampness on leakage.

Immediately after discharging the water off the slab, the top surface shall be examined for any water stagnation. Water collected beyond a depth of 3mm in any area is considered a stagnation.

After any epoxy pressure grouting rectification, hacking and refilling around pipes installed after the first test, the slabs shall be retested at the Contractor's own costs and expenses for watertightness, water stagnation or both as deemed necessary by the Engineer.

43.6 Floortraps / Scupper Drains

Floors of areas which contain floor traps and/or scupper drains shall have the surface of such floors sloping definitely towards the floor traps and/or drains.

43.7 Damp-Proof Membrane (Moisture Vapour Barrier)

All concrete slabs on ground level shall be laid on an approved damp-proof barrier consisting of 0.2mm thick polyethylene sheet.

Joints shall overlap a minimum of 150mm, and the edges taped.

For suspended-slab, the damp-proof membrane shall be in direct contact with the compacted ground and below the lean concrete.

For non-suspended slab (i.e. slab-on-grade), the damp-proof membrane shall be in direct contact with the compacted sub-base or subgrade and below the soffit of structural slab.

44.0 Slabs-On-Grade

44.1 Formation Level

Excavation to formation level shall not be over-cut by more than 100mm. Over-cut areas shall be filled with approved material.

Contractor shall prevent damage to the formation owing to exposure to the atmosphere and the trampling of the formation by workmen. Where the formation is softened or spoilt, the soft areas shall be excavated and filled with Grade 15 concrete at the Contractor's own costs and expenses.

44.2 Sub-Grade

The top 500mm of the sub-grade shall have a minimum CBR (California Bearing Ratio) value of 5% and shall be well compacted to minimum 95% of dry density.

44.3 Sub-Base

Unless otherwise stated in the drawing, the sub-base shall consist of inert well-graded granular material of maximum size 75mm, fully compacted and lightly blinded with sand or inert, crushed, fine material to form a flat surface to within a tolerance of +0mm, -25mm.

Sub-base shall be compacted to 95% of dry density and have a minimum CBR value of 30%.

44.4 Rate of Soil Testing (for slab-on-grade construction)

	Test to be carried out in accordance with BS 1377	Minimum frequency of testing
Sub-Grade	CBR test (5% min) In-situ field density : (min. 95% of dry density)	1 test per 800m ² or min. 4 tests, whichever the greater. 1 test per 400mm ² or min. 9 tests, whichever the greater.
Sub-Base	CBR test (30% min.) In-situ field density : (min. 95% dry density)	1 test per 800 m ² or min. 4 tests, whichever the greater 1 test per 400m ² or min. 9 tests, whichever the greater

44.5 Compaction

All compactions shall be carried out by 10 tonnes smooth wheel rollers or equivalent capacity vibrating roller.

44.6 Damp- Proof Membrane

A damp-proof membrane, 0.2mm thick polyethylene sheet, shall be laid in direct contact with the sub-base. Joints shall overlapped a minimum of 150mm and the edges taped.

44.7 Base Slab

Slab-on-grade shall be laid in strip. Alternate strips are constructed first, then infill strips are placed 2 or more days later, when the adjacent concrete has hardened.

Unless otherwise stated in the drawings, longitudinal (constructional) joints shall not be more than 4.5m apart and transverse (contraction) joints shall be provided in floor slabs at spacing not more than 6m. The sides of the joints shall be vertical and shall coincide with re-entrant corners.

For slabs constructed in the open, the joints shall be induced within 24 hours;

for those constructed under a roof, the time limit shall be extended to a maximum of 7 days after concreting.

Contraction joints shall be formed by sawing to a depth about $\frac{1}{4}$ of the slab thickness. Sawed joints in floor shall be sealed with approved sealant.

Unless otherwise stated in drawings, expansion and isolation joint shall be provided with approved 20mm thick closed cell expanded polyethylene foam joint compressible-filler and approved sealant.

The structural concrete work shall follow the provision of Clauses 1 to 40.

Power floating of the concrete shall be done after the bleed water is removed.

Floating should not commence until bleeding has stopped nor before the concrete is firm enough to permit a man to walk on with leaving impressions more than 3mm deep.

Power trowelling shall not be applied until the concrete surfaces has been floated.

45.0 Floor Hardener

45.1 General

When surface hardeners are required to be used, the Contractor shall submit for the Engineer's approval, the type, colour, composition, details of manufacture and the manufacturer's method of use. Any hardener consisting of soluble alkaline compounds, rust and materials designed to disguise rust shall not be allowed to be used. Upon approval of the type, the application of the hardener shall be as specified by the manufacturer.

The floor hardening work shall be carried out by an approved specialist.

The treated surface shall be non-dusting, non-rusting, non-magnetic, abrasion resistant, slip resistant, petrol, mineral oil and solvent resistant and impact resistant.

45.2 Abrasion Resistance Performance Criteria

When tested in accordance with the method of the Abrasion Resistance Test Method (SS73), the decrease in weight and thickness of the test sample after 1800 revolutions (after 60 minutes) shall not be greater than 17.5gm and 1.2mm respectively taken from an average of 3 test samples.

Submit abrasion resistance test report from SISIR accredited laboratory to the Engineer before commencement of work.

Provide core samples from the treated floor for abrasion resistance test when directed by the Engineer.

45.3 Application of Floor Hardeners

Place, compact and level concrete to the requirements of the Engineer. Approved floor hardener shall be applied, finished and cured in accordance with the Manufacturer's recommendations and instructions.

Submit complete details of the method of application of the floor hardener to the Engineer for approval before commencement of work.

45.4 Protection

Protect the floor against any damage until handing over to the Employer.

45.5 Warranty

The Contractor and the Specialist shall jointly warrant for a period of five (5)

years that the works shall be non-dusting, non-rusting, non-magnetic, abrasion resistant, slip resistant, petrol, mineral oil and solvent resistant, impact resistant and remain free from defects.

46.0 Roofs

46.1 General

Reinforced concrete roofs shall be constructed to fall as shown in the Drawings and finished with power trowelling, leaving the surface smooth and free from mortar droppings. The finished roof slab shall be cured immediately with an approved curing compound.

Roof openings, with 200mm upstanding kerbs, shall be constructed at the positions as shown in the Drawings.

100mm x 100mm mortar angle fillet shall be neatly provided around all upstands, pipe supports, RC stumps, walls and ventilating stacks.

The finished concrete roof shall be protected against mortar droppings and damages by heavy objects during subsequent work on the roof.

46.2 Watertightness and Water Stagnation Test

After the completion of all structural works on the roof, the roof slab shall be tested for watertightness and to ensure no water stagnation.

The watertightness test shall be carried out by ponding with water to a minimum depth of 25mm for 24 hours. At the end of the watertightness test, the soffit of the slab especially the area around the pipes shall be examined for any dampness or leakage.

Immediately after discharging the water off the roof, the top surface shall be examined for any water stagnation. Water collected beyond a depth of 3mm in any area is considered a stagnation.

After any epoxy pressure grouting, rectification, hacking and refilling around pipes installed after the first test, the localised area shall be retested at the Contractor's own costs and expenses for watertightness, water stagnation or both as deemed necessary by the Engineer.

47.0 Drips

Drips shall be formed along edges of soffit to concrete roof slabs, hoods, underside of balconies, cantilevered beams and slabs and other parts of building where rainwater is likely to adhere in drops.

48.0 Loads in Excess of Design Load

No load in excess of the design loading shall be placed on any portion of the structure without the written permission of Engineer. If such permission is granted, all beams or other members of the structure which are subjected to a load other than the designed load shall be strengthened and supported to the satisfaction of the Engineer at the Contractor's expense. The Contractor shall be responsible for making good to the satisfaction of the Engineer any damage to the permanent structure which may be caused by such excess loads.

49.0 Waterproofing

49.1 General

All waterproofing works shall be executed by a firm of specialists approved by the Engineer and the Contractor shall furnish a 10-year warranty for watertightness and against failures in respect of faulty workmanship and material. The warranty shall be issued jointly by the Contractor and the waterproofer.

The waterproofing shall be applied in strict conformation to the manufacturer's and supplier's recommendations and suggested details.

Contractor shall provide adequate measures to protect the waterproofing system from being damaged during subsequent constructional activities.

All joints of pipe penetrations/outlets through concrete structures at all wet areas and/or likely to be exposed to water shall be made waterproofed/leak proofed.

All pipes or other appurtenances that pierce through walls and floors shall have a minimum 50mm flange around it embedded in the concrete.

Where flanges are not provided, the openings allowed for such penetrations shall be filled with waterproof non-shrink grout.

Formwork to all rc walls below external ground level shall be provided with 'snap-tie' form-bolts. Recesses in concrete left by form-bolts shall be plugged with water-proof mortar.

49.2 Waterproofing Membrane at Wet Areas (including Toilets, Bath/WC, Wash Areas/Yards, Balconies, Kitchens and Flower Boxes)

The contractor shall provide and lay a high performance waterproofing system in the toilets, bath/wc, wash- areas/yards, balconies, kitchens and flower boxes.

The waterproofing system shall be applied over the entire floors of the toilets, bath/wc, wash- areas/yards, balconies, kitchens and flower boxes and shall be carried up the walls and pipes minimum 150mm from finish-level unless otherwise stated.

The waterproofing system shall be subject to the approval of the Consultants and no additional cost will be allowed.

All liquid applied membrane spanning across any joint shall be reinforced. The reinforcement used shall be compatible with the sandwiching membrane and

shall be an integrated proprietary system which comes with the membrane system.

The prepared surface to receive the membrane shall be free from dust, laitance and ground smooth if necessary. Provide for the use of primers where required.

As directed, apply and cure the membrane strictly in accordance with the recommendations and instructions of the Manufacturer and/or the Engineer.

After allowing for adequate curing, lay waterproof cement and sand screed to fall evenly towards water outlets. Provide adequate measures to protect the membrane system from being damaged during the screeding operations.

49.3 Waterproofing System to Flat Roofs

The Contractor shall provide and lay a complete waterproofed roofing system as specified in the Architectural drawings and specification.

49.4 Waterproofing Admixture for Reinforced Concrete Roof

The application shall be in accordance with the manufacturer's recommendations and instructions.

Contractor shall check and ensure that the admixture does not affect the strength or other properties of the reinforced concrete.

The admixture shall comply with SS EN 934 and shall be approved by the Engineer.

49.5 Treatment of Pipe Leads-In

All pipes or other appurtenances that pierce through walls and floors shall have a minimum 50mm flange around it embedded in the concrete.

50.0 WATERTIGHTNESS TEST

All reinforced concrete structures designed to retain water or chemical solutions are to be tested for watertightness to the complete satisfaction of the Engineer. The test includes providing all necessary blank flanges, temporary plumbing, pumps and water and all other requisite materials and plant.

The Contractor shall test all aqueous liquid retaining structures, (including swimming pools, spa pools, balancing tanks, etc) for watertightness in accordance with the latest SS EN 1992 - 3 requirements 28-days after casting and before the application of waterproofing material.

The aqueous liquid retaining structure shall be cleaned and filled with PUB water gradually up to overflow level at a uniform rate of not greater than 2m in 24 hours and tested for possible leakage.

The period of test shall be at least 7 days during which the exposed surfaces shall show no sign of leakage, no patches of dampness occur and remain apparently dry.

During the 7-day test period the total permissible drop in level, after allowing for evaporation and rainfall, shall not exceed 1/500th of the average water depth of the full tank.

If the aqueous liquid retaining structure leaks or shows sign of leakage, it shall be repaired with approved type of Epoxy resin pressure grout and the structure be retested.

If the aqueous liquid retaining structure fails the second test, the Engineer reserves the right to instruct the Contractor to remove and recast the structure at his own costs and expenses. The recast structure shall be required to pass the same watertightness test.

All watertightness tests shall be witnessed and verified by the Engineer.

51.0 REINFORCED CONCRETE AQUEOUS-LIQUID RETAINING STRUCTURES: INCLUDING SWIMMING POOLS, SPA POOLS AND BALANCING TANKS

The construction of all reinforced concrete aqueous- liquid retaining structures shall be in accordance with the latest SS EN 1992 – 3 or local code requirements.

All positions and sizes of openings, sumps, drains etc. to be confirmed with the Architect and M&E Engineer before commencement of works.

All pipes or other appuntenances that pierce through walls and slabs shall have a minimum 50mm flange around it embedded in the concrete.

51.1 Structural Concrete

Grade 35 concrete shall be used for the construction of the concrete aqueous-liquid retaining structures.

The concrete mix shall be designed in accordance with Clause 6 'Concrete Mixes' and shall be approved by the Engineer.

The concrete shall be mixed with approved type of chloride-free retarding super plasticiser to achieve a highly workable mix with maximum water/cement ratio of 0.5.

The proportion and dosage required for the admixture shall be carefully designed and applied in accordance with the Manufacturer's instructions to achieve a high quality concrete.

All defective concrete work such as honeycomb, surface air pockets, poor alignment, level, finishes, etc shall be reported to the Engineer and rectified to the instruction of the Engineer, all at the costs and expenses of the Contractor.

51.2 Formwork System

The location and method of installation of form ties and/or fixing device shall be approved by the Engineer and shall not remain within specific concrete cover.

Formwork to walls shall be provided with approved type of snap-tie form ties. Recesses in concrete left by form-ties shall be plugged with high strength non-shrink waterproof mortar.

Formwork shall only be removed at least 7 days after concreting of structure and the soffit of elevated- structure be reproped immediately until 28 days after casting.

Formwork shall be removed gradually without shock, disturbance and vibration, without damaging or overloading the finished concrete work or structures, and in manner approved by the Engineer.

Unless otherwise specified, formwork shall be constructed generally in accordance with Clause 23 and Clause 27 where appropriate.

51.3 Concreting

All construction joints and concreting procedures proposed by Contractor for the aqueous liquid retaining structure work shall be approved by the Engineer.

Contractor shall inform the Engineer the areas of concreting and the locations of construction joints and obtain his approval prior to concreting.

For structures below ground level, waterstops shall be provided for all horizontal and all vertical construction joints. Rearguard PVC waterstops 250mm wide x 3mm thick shall be provided for floor joints and vertical walls joints and kicker joints.

The cost of providing the waterstops is deemed to be included in the Contractor's price, excluding those indicated on drawings.

The concrete shall be highly workable and shall be compacted sufficiently to produce a dense homogeneous mass and filling all voids. The concrete shall not be subjected to any disturbance after being placed in position and shall be continuously cured for a minimum of 7 days after casting.

51.4 51.4 Water Tightness Test

All aqueous liquid retaining structures shall be tested for watertightness in accordance with Clause 47.

The Contractor shall test the structure for watertightness in accordance with SS EN 1992 - 3 requirements 28-days after casting and before the application of waterproofing material.

51.5 Waterproofing

The waterproofing shall be applied after satisfactory completion of watertightness test.

Unless otherwise specified, waterproofing work shall be executed in accordance with Clause 46.

51.6 Backfilling

For structures below ground level, backfill excavation with sand after completion of waterproofing.

Contractor shall provide and fix 25mm thick, grade 16kg/m³ polystyrene protection board to external wall surfaces of concrete structure below ground level before backfilling.

52.0 LIGHTWEIGHT CONCRETE

Where lightweight concrete is indicated, the Contractor shall submit details of his proposed mix and resulting density of the concrete for approval. Lightweight concrete shall be mixed and poured strictly in accordance with the instructions of the manufacturer of the approved lightweight aggregate used.

Lightweight concrete shall comply with the requirements of EN 13055 or equivalent codes.

53.0 PRECAST CONCRETE

53.1 General

The concrete in one precast shall be placed in one operation. No pieces shall be removed from the mould or erected until sufficiently matured to ensure that no damage shall be done to the piece. A piece shall be suspended or supported only at the points described on the working drawings or elsewhere. A piece that is cracked or otherwise damaged during, before or after erection shall be removed from the works and replaced by the Contractor at his own expense. Pieces shall be bedded or otherwise fixed in their permanent positions as instructed.

53.2 Precast Concrete Slab

All precast concrete and revetment slabs shall satisfy the requirements of EN 1340.

(a) Freedom from Defects

The thickness of each slab shall be uniform and all angles shall be true right angles. The edges shall be clean and sharp and the surfaces shall be true and free from any winding. On being fractured, the interior of the slabs shall present a clean homogeneous appearance.

(b) Transverse Strength

When tested in the manner described in EN 1340 or local code requirements, the transverse strength shall comply with EN 1340.

(c) Absorption of Water

When tested in the manner described in EN 1340 or local code requirements, the average increase in weight of the test pieces from each slab by absorption of water shall not exceed the appropriate values in EN 1340.

53.3 Precast Concrete Pipes

All precast concrete pipes shall be centrifugally spun reinforced concrete pipes of a minimum standard equivalent to Standard Class. All pipes shall conform to SS 183 or equivalent code.

53.4 Precast Kerbs and Channels

All precast concrete kerbs and channels shall conform entirely to the shape and dimensions shown on the drawings and shall satisfy the requirements of SS 214 or equivalent code.

(a) Freedom from Defects

All angles of the products, with the exception of the angles resulting from the splayed or chamfered fences, shall be true right angles. The arrises shall be cleaned and with the exception of rounded arrises, sharp. The wearing surface shall be true and free from any winding. On being fractured, the interior of the products shall present a clean homogeneous appearance.

(b) Transverse Strength

When tested in accordance with SS 214 or equivalent code, the transverse strength shall comply with SS 214 or local code requirements.

(c) Absorption of Water

When tested in accordance with SS 214 or equivalent code, none of the three specimen absorptions shall exceed the appropriate values in SS 214 or local code requirements.

53.5 Testing of Precast Units of Slab, Kerbs and Channels

The Contractor shall, at his own expenses, supply all necessary labour, implements and transport required for the testing of the precast units and deliver the test samples from the site to an approved Accredited Laboratory for testing as and when and in the numbers required by the Engineer. The Contractor shall meet all expenses incurred in the testing of precast units.

Samples shall be taken at random by the Engineer from the appropriate consignment representing the whole or part of the order. Three samples precast units shall be taken for every 2000 units or less. For orders of more than 2000 units, three samples units shall be taken from every 2000 units forming part of the order. Should any of the test results not comply with requirements, then the whole of the consignment of 2000 units, or less, comprising the whole or part of the order, shall be deemed not to comply with this Specification and liable to rejection by the Engineer.

53.6 Precast Sills and Copings

All precast concrete sills and copings shall conform to the shape and dimension shown on the Architectural drawings and shall satisfy the requirements of BS 5642 or local code requirements.

Reinforcement provided shall be a section of long mesh fabric (C6) complying with the requirements for Grade C283 in SS561 or local code requirements and cut to provide two longitudinal bars.

53.7 Precast Concrete Building Blocks

Precast concrete building blocks, either of an approved proprietary brand or cast by the Contractor, shall comply with the requirements of EN 771 or local code requirements.

The Engineer may select random blocks from stocks at the site for delivery to an approved Accredited Laboratory for testing purposes. Should blocks subjected to test not attain the required strengths, the Engineer may order that the remaining blocks be removed from the site.

53.8 Precast Lintels

All precast concrete lintels shall satisfy the requirements of EN 845 or local code requirements.

The end bearing for lintels shall be not less than 150mm.

54.0 SCHEDULE OF CONSTRUCTION TESTS FOR CONCRETE WORKS

The Contractor shall allow and conduct all construction tests as prescribed in the local or international code requirements.

Such tests shall be carried out by an Accredited Laboratory.

Tests to be carried out	Standard to be used	Minimum frequency of testing/the numbers of testing required
1) Cement	SS EN 197-1	
2) Aggregate	BS 812 for compliance with SS EN 12620	To be obtained from concrete supplier
3) Water	EN 1008	
4) Concrete Cubes	EN 12350, EN 12390, EN 12504 and BS 1881	1 sample of 4 cubes per 40m ³
5) Steel Reinforcements: • Tensile Test • Bend Test and • Rebend test	SS 560 SS 561	1 sample of 3 test pieces per size per consignment

55.0 Document Control

55.1 Approval: GLOBAL_FAC_GFTT_APPR

55.2 Notification:

GLOBAL_FAC_MANAGERS

GLOBAL_FAC_COMMUNICATIONS

GLOBAL_FAC_CONSTRUCTION

GLOBAL_FAC_NOTIFY

FCG_F4_FAC_NOTIFY

FCG_F6_FAC_NOTIFY

FCG_F10_FAC_NOTIFY

FCG_F15_FAC_NOTIFY

FCG_F16_FAC_NOTIFY

FCG_MMP_FAC_NOTIFY

FCG_MMY_FAC_NOTIFY

FCG_MSB_FAC_NOTIFY

FCG_MXA_FAC_NOTIFY

FCG_MTB_FAC_NOTIFY

55.3 Notification:

All Micron Team Members at all Sites

55.4 Retention

Adherence to the requirements specified in

[Global Facilities - Document Control Procedures](#)

54.5 Review

Biennially (two years)

56.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New Document	07/04/2022